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Challenges set 1: UDP analysis

Sample to use: DNS+NTP

1. Open the DNS sample
2. Locate the UDP header
3. Write down the source port and destination port
4. What is the destination port number and which well-known service does it represent?

Source port: 59198

Destination port: 54

192.168.56.11 → 194.190.124.17 DNS Standard query 0xe4a3 A ns2.highsecure.ru OPT

[Ask AI about this packet](#)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
52	54	00	7E	42	A5	52	54	00	A6	11	53	08	00	45	00
00	4A	4A	48	00	00	3F	11	F9	D7	C0	A8	38	0B	C2	BE
7C	11	E7	3E	00	35	00	36	37	CB	E4	A3	00	10	00	01
00	00	00	00	00	01	03	6E	73	32	0A	68	69	67	68	73
65	63	75	72	65	02	72	75	00	00	01	00	01	00	00	29
10	00	00	00	80	00	00	00								

4 protocols in packet (88 bytes):

Ethernet

IPv4

UDP

DNS

The image shows a Wireshark packet capture interface. At the top, there are buttons for '+', 'Filter tree...', 'Width: 16 bytes', 'Export', and 'Share'. Below these, the packet list on the left shows a single packet: 'Frame 1: 88 bytes on wire (704 bits)'. The packet details pane on the right shows the following structure:

- * **Frame 1: 88 bytes on wire (704 bits)**
- * **Ethernet II**
- * **Internet Protocol Version 4**
- * **User Datagram Protocol**
 - * Source Port: 59198
 - * Destination Port: 53
 - * Length: 54
 - * **Checksum: 0x37cb [correct] (matches partial checksum, not 0x647a, likely caused by "UDP checksum offload")**
 - * Checksum Status: Good
 - * Stream index: 0
 - * Stream Packet Number: 1
 - * **Timestamps**
 - * UDP payload (46 bytes)
- * **Domain Name System (query)**

2. STILL IN THE DNS SAMPLE:

UDP length: 54

UDP checksum: 0x37CB

Sequence number: not present

Acknowledgement number: not present

conclusion: UDP does not use sequence numbers or acknowledgement numbers. This shows that UDP is a connectionless protocol and does not provide TCP-style reliability mechanisms.

3. Open the NTP Sample:

The UDP headers of DNS and NTP contain the same four fields: source port, destination port, length, checksum

The UDP Header has a fixed size of 8 bytes

UDP is suitable for both DNS & NTP because it has low overhead and provides fast, connectionless communication. Both applications can use UDP port numbers to identify the required services.

Decode

Upload

Clear

Samples ▾

216.27.185.42 → 192.168.50.50 NTP NTP Version 3, symmetric passive

+ Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	D0	59	6C	40	4E	00	0C	41	82	B2	53	08	00	45	00
00	4C	A2	22	40	00	32	11	22	5E	D8	1B	B9	2A	C0	A8
32	32	00	7B	00	7B	00	38	70	E1	1A	02	0A	EE	00	00
07	A4	00	00	0B	A3	A4	43	3E	C2	C5	02	01	81	E5	79
18	19	C5	02	04	EC	EC	42	EE	92	C5	02	04	EB	D9	5E
8D	54	C5	02	04	EB	D9	69	B1	74						

4 protocols in packet (90 bytes):

Ethernet

IPv4

UDP

NTP

2. Challenges set 2:

Decode

Upload

Clear

Samples ▾

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231]

+ Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
46	54	50	44	20	41	6E	6F	6E	79	6D	6F	75	73	20	53
65	72	76	65	72	29	20	5B	31	39	32	2E	31	36	38	2E
31	2E	32	33	31	5D	00	0A								

4 protocols in packet (136 bytes):

Ethernet

IPv4

TCP

FTP

+

Filter tree...

width: 16 bytes ▾

Export ▾

Share

- ▶ **Frame 1: 136 bytes on wire (1088 bits)**
- ▶ **Ethernet II**
- ▶ **Internet Protocol Version 4**
- ▶ **Transmission Control Protocol**
 - Source Port: 21
 - Destination Port: 62014
 - Stream index: 0
 - Stream Packet Number: 1
- ▶ **Conversation completeness: Incomplete (0)**
 - TCP Segment Len: 70
 - Sequence Number: 0x836f4ffe
 - Sequence Number (raw): 2205110270
 - Next Sequence Number: 71
 - Acknowledgment Number: 0x04846eda
 - Acknowledgment number (raw): 75787994
 - 1000 = Header Length: 32 bytes (8)
- ▶ **Flags: 0x018 (PSH, ACK)**
 - Window: 1448
 - Calculated window size: 1448
 - Window size scaling factor: -1 (unknown)

What is 4-tuple that uniquely identifies this TCP socket?

The 4-tuple is

192.168.1.231,21,192.168.1.182,62014

The tuple consists of the source IP address,source port,destination IP address,and destination port. It uniquely identifies a TCP connection and helps the operating system perform demultiplexing by delivering each incoming packet to the correct TCP socket/application

▸ Internet Protocol Version 4

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- **Differentiated Services Field: 0x10 (DSCP: Unknown, ECN: Not-ECT)**
- Total Length: 122
- Identification: 0xe65e (58974)
- **010. = Flags: 0x2, Don't fragment**
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: TCP (6)
- Header Checksum: 0xcf21 [correct]
- Header checksum status: Good
- Calculated Checksum: 0xcf21
- Source Address: 192.168.1.231
- Destination Address: 192.168.1.182
- Stream index: 0

6. Open mysql sample:

192.168.0.254 → 192.168.0.254 MySQL Request Query [Ask AI about this packet](#)

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	00	00	00	00	00	00	00	00	00	00	00	08	00	45	08
00	4A	65	C9	40	00	40	06	51	90	C0	A8	00	FE	C0	A8
00	FE	DB	62	0C	EA	CC	D8	BB	B5	CD	34	96	7B	80	18
02	01	83	89	00	00	01	01	08	0A	00	F0	F4	D1	00	F0
DE	8F	12	00	00	00	03	53	45	4C	45	43	54	20	44	41
54	41	42	41	53	45	28	29								

4 protocols in packet (88 bytes):

Ethernet IPv4 TCP MySQL

FTP destination:662014

MySQL destination:3306

The destination ports are different.

The operating systems uses the IP address and port numbers of each TCP connection to perform demultiplexing.this allows it to allows to identify the correct and socket and deliver each packet to the appropriate application.

Challenges set 3: TCP vs UDP comparison

TCP has a larger header than UDP. The minimum TCP header size is 20 bytes, while the UDP header is 8 bytes.

TCP provides reliability features such as sequence numbers, acknowledgement numbers, flags, and window size. UDP does not provide sequence numbers or acknowledgement numbers

. TCP is connection-oriented, while UDP is connectionless. The FTP sample shows TCP sequence and acknowledgement numbers, whereas the DNS sample using UDP does not contain these fields.

8. From the FTP sample: o Point out any fields that show TCP's duty of providing reliability and flow control.

The FTP TCP header contains several fields that provide reliability and flow control:

- Sequence Number: 0x836F4FFE — helps maintain the correct order of data.
- Acknowledgement Number: 0x04846EDA — acknowledges received data.
- ACK Flag: Set — indicates that acknowledgement information is present.
- Window Size: 1448 — provides flow control by indicating the amount of data the receiver can accept

.

Therefore, Sequence Number and Acknowledgement Number support reliability, while Window Size provides flow control.



Challenges set 4: modern transport

Sample to use:QUIC

Underlying Transport Protocol:

UDP Destination Port:443

QUIC runs over UDP but provides its own transport features, including reliability and connection management. Unlike classic TCP, QUIC does not use the TCP protocol and header; it uses UDP as its underlying transport.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	03	2D	87	59	70	F0	18	98	E8	7A	F3	86	DD	60	03
0E	00	00	3D	11	40	20	01	0D	B8	00	01	00	00	00	00
00	00	00	00	00	01	26	06	47	00	00	10	00	00	00	00
00	00	68	16	08	26	C4	68	01	BB	00	3D	F6	CF	E4	00
00	00	01	14	01	30	DF	C5	A0	47	E6	AC	D2	30	B5	C5
E0	47	CE	D9	B0	A6	BB	F0	00	40	18	62	62	7A	4B	50
9A	40	68	12	B7	1D	F9	BE	5A	8C	E4	F8	95	DD	26	6E
54	56	7D													

4 protocols in packet (115 bytes):

Ethernet

IPv6

UDP

QUIC

User Datagram Protocol

Destination Port: 443

Field offset: [56-57]

Field length: 2

Byte offset: 56